

Project Brief



FIFA World Cup 2026

All the terrific matches of the FIFA World Cup 2026 have just ended. Advances in technology and football analytics leave a fascinating trove of team-level and player-level data from the following websites:

- FIFA Official Website: <https://www.fifa.com/en/tournaments/mens/worldcup/canadamexicousa2026/statistics> (<https://www.fifa.com/en/tournaments/mens/worldcup/canadamexicousa2026/statistics>)
- The Stats Don't Lie: <https://www.thestatsdontlie.com/football/world-cup-2026/> (<https://www.thestatsdontlie.com/football/world-cup-2026/>)
- FB Ref: <https://fbref.com/en/> (<https://fbref.com/en/>)

Objective 1

Your team will perform four (4) distinct analytic tasks that analyse any of the FIFA World Cup 2026 datasets above using Python. You are free to decide the specific tasks to carry out, as long as each task adheres to the requirements below.

*Each analytic task **must** be driven by a distinct question that will be answered using **all** the following skills:*

- Analytic question formulation

- Data wrangling
- Data preparation and sampling
- Descriptive statistics
- Inferential statistics (Confidence interval)
- Inferential statistics (either a One-Sample t-Test OR a Two-sample t-Test)

Each analytic task **must also** have a distinct focal point of investigation that differs from the other tasks. For instance, two tasks that focus on analysing the number of goals are not considered distinct and will only be considered as one task.

Example of an analytic task.

Note: You CAN NO LONGER use this same analytic task in your assignment.

- Analytic question formulation: On average, how many goals were scored by a forward player during the FIFA World Cup 2026? Exclude players who have never appeared in any match.
- Data wrangling: Obtain players' details and match history during this World Cup. Filter out those who never played in any match.
- Data preparation and sampling: Collect appropriate variables. Define population vs sample. Obtain a sufficient sample using an appropriate technique.
- Descriptive statistics: Analyse the sample using appropriate descriptive statistics.
- Inferential statistics (Confidence interval): Use the sample statistics to obtain the confidence interval estimation for the player population.
- Inferential statistics (either a One-Sample t-Test OR a Two-sample t-Test): Meaningfully compare the average goals of two groups of players.

Objective 2

Your team will perform two related but distinct linear regression tasks as specified below.

Linear Regression 2.1

Build and evaluate a linear regression model that predicts the goal difference between two opposing teams in FIFA World Cup 2026 matches. You must compile a dataset that comprises 104 rows, each representing an actual match in the FIFA World Cup 2026. Your dataset must include **exactly** eight (8) explanatory variables (excluding any ID field). You must thoughtfully decide what these explanatory variables are and collect their data.

Note:

- All explanatory variables **must** provide information that is available *before* a match. For example, the number of shots on target made during a match cannot be used as an explanatory variable for predicting the goal difference of *that* match.

Linear Regression 2.2

Build and evaluate a second linear regression model that predicts the number of goals scored by a team in a single FIFA World Cup 2026 match. You must compile a dataset that comprises 208 rows, each representing a team's statistics for a specific match. Your dataset must include exactly eight (8) explanatory

variables (excluding any ID field). You must thoughtfully decide what these explanatory variables are and collect their data.

Note:

- All explanatory variables **must** provide information that is available *before* a match. For example, the number of shots on target made by a team during a match cannot be used as an explanatory variable for predicting their goal for *that* match.

Trouble viewing this page? Go to our diagnostics page to see what's wrong.
Skip to main content



VIBHAGH
student
20 days ago
2 at 10:00 PM

It says in the assignment that we have to submit the assignment in a group project presentation. Does that mean I need to submit my assignment in a group? I need to submit my assignment in a group. Is the final version of the slide or file at once?

YAKUB SEBASTIAN

+ 43H

⋮

👍 2

😊

🔍

1

💬

1This post has 1 comment



Post

Yakub Sebastian





JEFFREY
student
6 days ago
1 at 10:00 PM

Hi Yakub, I am a student. For the assignment, we have to perform a task. Will we need a confidence interval task?

⋮

👍 1

😊

💬 1This post has 1 comment